

Counting events

Poisson, negative binomial and hurdle models

Silje Synnøve Lyder Hermansen

The dependent variable

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Count models: What are they good for?

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⇒ *e.g. number of meetings between decision makers, violent events, legislative proposals, etc.*

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⇒ Variables are on the exposure level; related to when (where) the events took place.

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⇒ *We replace the normal distribution with another probability distribution*

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Formula

The equation the model estimates:

$$E(y_i) \equiv h\lambda_i = h \times \exp(\alpha + \beta \times x_i) \quad (1)$$

Estimation of the exposure

What to do with the exposure parameter?

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Two strategies :

- **Offset:** Move it into the equation but constrain parameter:
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- ▶ **Estimate a parameter:** $\exp(\alpha + \beta_1 \times x_i + \beta_2 \times \log(h_i)) \rightarrow$ *when exposure is different*

$\Rightarrow$ *If the exposure is the same for all units, we set it to 1 and ignore it (R does that).*

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 - ▶ Marginal effect of β : $\exp(\beta)$ is multiplicative of predicted $\hat{\lambda} \rightarrow$ easy!

$\Rightarrow$ *Make scenarios, predict, knock yourself out*

Dispersion

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$\Rightarrow$ *The standard errors will be too small*

Identifying overdispersion

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- ▶ Events are related

Addressing overdispersion

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$\Rightarrow \beta$ *remains the same, standard errors are larger*

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⇒ We can model this in two parallel regressions with possibly different x or just an additional intercept.

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⇒ *Can accomodate under-dispersion too.*

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⇒ functions as a switch that is turned on/off after a threshold. The observation is then passed to the count-model group.

Recap on GLMs

What are the criteria for model selection?

You can think of model selection as a set of criteria that should be met

Try out the model selection decision tree to see my mental map!

https://siljehermannsen.github.io/teaching/choose_glm/